

Statistics Practice Workbook

A Synthetic Dataset + 26 Hands-On Exercises Across Every Major Test

Prepared for Irfan Ansari | Run every test yourself in SQL/Python, then check against the answer key

What's in This Kit

This kit gives you a realistic, engineered dataset and a matching set of exercises so you can actually *run* every major statistical test yourself instead of just reading about them — the fastest way to make these concepts stick before an interview.

File	What it's for
game_analytics_practice_dataset.csv	2,500 synthetic mobile-game users with realistic, engineered relationships (an A/B test, platform effects, skewed revenue with outliers, etc.) — the main dataset for Q1–21 and Q25–26.
simpsons_paradox_practice_dataset.csv	A small aggregated table of two onboarding flows across two user segments, engineered to reproduce Simpson's Paradox — used for Q22–24.
practice_starter.py	A Python starter script with the dataset already loaded and one stubbed section per question — fill in the code yourself, then compare your output to the Answer Key at the back of this document.

How to use this: Try to answer each question cold first — pick the right test before you even open Python. Then write the code, run it, and only after that check the Answer Key (Section 9) to confirm your number and your conclusion both came out right. Getting the right number with the wrong test picked "by luck" is the failure mode this is meant to catch.

Data Dictionary

game_analytics_practice_dataset.csv (2,500 rows, 1 row per user)

Column	Type	Description
user_id	int	Unique user identifier
platform	categorical	iOS or Android
country	categorical	US, UK, India, Brazil, Germany
marketing_channel	categorical	Organic, Paid Social, Paid Search, Referral
ab_group	categorical	Control or Treatment — a randomized 50/50 feature test
signup_week	int (1–12)	Week of signup within a 12-week acquisition window
age	int	User age in years
session_count_30d	int	Number of sessions in the first 30 days
avg_session_length_min	float	Average session length in minutes
retained_d7	binary (0/1)	Active again on/after day 7
retained_d30	binary (0/1)	Active again on/after day 30
nps_score	int (0–10)	Self-reported Net Promoter Score
purchases_30d	int	Number of in-app purchases in 30 days
revenue_30d	float	Total revenue in 30 days (USD) — right-skewed with a small number of "whale" outliers

simpsons_paradox_practice_dataset.csv (4 rows, aggregated)

Column	Description
segment	New User or Power User
flow	Onboarding flow A or B
converted	Number of users who converted
total	Number of users shown that flow in that segment
conversion_rate	converted / total

Section 1 · Descriptive Statistics & Distributions

Q1. Compute the mean and median of `revenue_30d`. Which one is more representative of a "typical" user, and why?

Q2. Compute the skewness of `revenue_30d`. What does the value tell you about the shape of the distribution, and why does that happen with revenue data specifically?

Hint: look at how many users have zero purchases.

Q3. Using the IQR method ($1.5 \times \text{IQR}$ beyond Q1/Q3), compute the outlier bounds for `revenue_30d` and count how many rows are flagged as outliers.

Q4. Construct a 95% confidence interval for mean `revenue_30d`. Write one sentence interpreting it correctly (not as "95% of users fall in this range").

Section 2 · One-Sample & Two-Sample Tests

Q5. Leadership claims the average user has 9.0 sessions in their first 30 days. Test whether the actual mean `session_count_30d` is significantly different from 9.0.

One-sample t-test.

Q6. Test whether `avg_session_length_min` differs between iOS and Android users. State the assumption you'd normally check (equal variances) and whether you used Welch's correction.

Independent two-sample t-test.

Q7. Test whether `revenue_30d` differs between the Treatment and Control groups using a standard t-test. Given what you found about this column's shape in Q2, is a t-test on the raw values fully appropriate? Re-run the comparison with a Mann-Whitney U test and see if your conclusion changes.

Two-sample t-test vs. non-parametric alternative.

Q8. `retained_d7` and `retained_d30` are two measurements on the *same* users, not independent groups. Which test family would you reach for to compare paired/related binary outcomes like this (you don't need to run it — name it and explain why a standard chi-square test of independence would be the wrong tool here).

Section 3 · Proportions & Categorical Tests

Q9. This is the core experiment in the dataset. Run a two-proportion z-test comparing `retained_d7` between the Control and Treatment groups. Report the absolute lift, the relative lift, and the p-value, and state your conclusion.

Q10. Build a 95% confidence interval for each group's D7 retention rate from Q9. Do the intervals overlap? What would overlapping intervals have implied?

Q11. Run a chi-square test of independence between `platform` and `retained_d7`. Is retention associated with platform?

Q12. Run a chi-square goodness-of-fit test to check whether `marketing_channel` volumes are evenly split across the four channels (25% each).

Section 4 · ANOVA

Q13. Run a one-way ANOVA testing whether mean `revenue_30d` differs across the four `marketing_channel` groups. What's your conclusion at $\alpha=0.05$?

Q14. Run a one-way ANOVA testing whether mean `session_count_30d` differs across `country`. Why is a clean "not significant" result here just as useful to know how to report as a significant one?

Section 5 · Correlation & Regression

Q15. Compute the Pearson correlation between `age` and `session_count_30d`. Even if it's statistically significant, is it practically meaningful? Why might a large dataset make this distinction especially important?

Q16. Compute the Spearman correlation between `nps_score` and `revenue_30d`. Why is Spearman arguably a better choice than Pearson for this specific pair of columns?

Q17. Fit a linear regression: `revenue_30d ~ session_count_30d + age + nps_score`. Report the R^2 and interpret the `nps_score` coefficient in plain language.

Q18. The R^2 from Q17 is very low even though `nps_score` is statistically significant. Explain how both can be true at once, and suggest a modeling approach better suited to a revenue column like this one (hint: think about what's driving the shape you found in Q2).

Q19. Fit a logistic regression: `retained_d7 ~ session_count_30d + age + ab_group + platform`. Convert the `ab_group` coefficient to an odds ratio and interpret it in one sentence.

Section 6 · Power & Sample Size

Q20. Using the retention rates from Q9, calculate the sample size per group needed to detect that same effect at 80% power and $\alpha=0.05$. Compare it to the actual group sizes in this dataset — was this experiment adequately powered?

Q21. If the true lift had been half as large, would the required sample size double, more than double, or less than double? Reason about the direction using the sample-size formula rather than recomputing exactly.

Section 7 · Simpson's Paradox Drill

Q22. Using `simpsons_paradox_practice_dataset.csv`, compute the conversion rate of Flow A vs. Flow B within each segment, then aggregate across segments to get the overall rate for each flow. What do you find?

Q23. Explain in your own words why the aggregate result reverses relative to the segment-level results. Name the concept.

Q24. If a product manager asked you "which flow should we ship," what would you actually recommend, and what additional analysis (e.g., a segment-weighted or stratified comparison) would you propose instead of just quoting the overall numbers?

Section 8 · Putting It Together

Q25. Write a two-paragraph stakeholder summary of this "experiment" as if presenting to a product manager, combining: the retention lift from Q9, the weak revenue regression from Q17–18, and the platform/retention association from Q11. Keep it in plain language with the key numbers, not the statistical machinery.

Q26. Identify at least two other columns in the dataset that could plausibly differ between the Treatment and Control groups purely by chance (not because of the treatment). How would you check that random assignment actually balanced these covariates before trusting the Q9 result?

Section 9 · Answer Key

Computed directly from the dataset with `scipy/statsmodels`. Use this to check your number *and* your test choice — not to skip running the code yourself.

A1/A2 (Q1-2). mean = 5.72, median = 0.00. The median is more representative — over half of users made \$0 in purchases, and a handful of "whale" spenders pull the mean far above what a typical user actually spent. Skewness ≈ 13.0 — extremely right-skewed, driven by the $\sim 50\%$ of users with zero revenue plus rare high-spend outliers.

A3 (Q3). Q1=0.00, Q3=7.38, IQR=7.38 $\rightarrow$ outlier bounds = (-11.07, 18.44). 172 of 2,500 rows ($\approx 6.9\%$) flagged as outliers (all on the high side, since revenue can't be negative).

A4 (Q4). Mean = 5.715, 95% CI $\approx$ (5.00, 6.43). Correct interpretation: "if we repeated this sampling process many times, 95% of the intervals constructed this way would contain the true population mean revenue" — not "95% of users earn between \$5.00 and \$6.43."

Q5. Sample mean `session_count_30d` = 9.99. One-sample t-test vs. 9.0: $t=13.44$, $p<0.0001$ — reject H_0 ; the true mean is significantly different from (higher than) 9.0.

Q6. iOS mean=14.94 min ($n=1,136$), Android mean=12.70 min ($n=1,364$). Welch's $t=7.97$, $p<0.0001$ — significant difference; Welch's correction is the safe default since group sizes and variances aren't guaranteed equal.

Q7. Standard t-test: Treatment mean=5.84 vs Control mean=5.59, $t=0.35$, $p=0.728$ — not significant. Mann-Whitney U (medians both 0.00): $p=0.692$ — also not significant. Both tests agree here: this heavily zero-inflated data made the t-test's normality violation less consequential for this particular comparison, but the non-parametric check is still the right instinct to reach for.

Q8. McNemar's test — designed for paired binary outcomes on the same subjects (before/after or two related yes/no measurements). A chi-square test of independence assumes independent observations, which day-7 and day-30 status for the same user are not.

Q9 (the core A/B result). Treatment D7 retention = 44.91% ($n=1,287$); Control D7 retention = 34.21% ($n=1,213$). Absolute lift = +10.70 points; relative lift = +31.3%. $z=5.46$, $p<0.0001$ — highly significant. Conclusion: the treatment meaningfully improves D7 retention.

Q10. 95% CI Treatment $\approx$ (42.2%, 47.6%); 95% CI Control $\approx$ (31.6%, 36.9%). The intervals do not overlap, reinforcing the significant result from Q9 — overlapping CIs would have suggested (though not proven) the difference could be due to chance.

Q11. Chi-square = 5.39, $\text{dof}=1$, $p=0.0203$ — significant at $\alpha=0.05$. Retention is associated with platform (iOS users retain somewhat better), consistent with the platform effect built into this dataset.

Q12. Observed counts: Organic 1,019, Paid Social 629, Paid Search 499, Referral 353 (vs. 625 expected each). Chi-square=392.2, $p<0.0001$ — channel mix is far from uniform.

Q13. Channel means: Organic \$5.18, Paid Search \$6.27, Paid Social \$4.99, Referral \$7.76. $F=2.26$, $p=0.080$ — not significant at the conventional 0.05 threshold, though Referral's higher mean is suggestive and could be worth a larger sample or a targeted follow-up test.

Q14. Country means all cluster around 9.8–10.1 sessions. $F=0.29$, $p=0.885$ — no significant difference. Reporting a clean null result correctly (rather than fishing for significance elsewhere) is exactly the discipline interviewers are checking for.

Q15. $r = -0.112$, $p<0.0001$. Statistically significant (large sample makes even small correlations detectable) but practically tiny — age explains only about 1.25% of the variance in session count ($r^2 \approx 0.0125$). With $n=2,500$, always check effect size alongside significance.

Q16. Spearman $\rho = 0.153$, $p<0.0001$. Spearman is preferable because `revenue_30d` is heavily skewed with many zeros/outliers (Q2), and `nps_score` is ordinal — Spearman's rank-based approach is robust to both issues, where Pearson would be distorted by the extreme values.

Q17. $R^2 = 0.008$ (essentially no explanatory power). `nps_score` coefficient ≈ 0.79 ($p<0.001$): holding session count and age constant, each additional NPS point is associated with about \$0.79 more revenue on average — statistically real, but the model overall barely explains any of the variation in revenue.

Q18. A predictor can be statistically significant while R^2 stays low when the outcome has enormous unexplained variance elsewhere — here, revenue is zero-inflated (about half of users spend nothing) and a handful of whales create huge unpredictable swings that no set of these three predictors can capture. Better approaches: a two-part/hurdle model (first predict "does this user purchase at all" with logistic regression, then model spend amount only among purchasers), or log-transforming revenue before regressing.

Q19. $\text{ab_treat coefficient} = 0.290 \rightarrow \text{odds ratio} = e^{0.290} \approx 1.34$. Interpretation: holding session count, age, and platform constant, being in the Treatment group is associated with about 34% higher odds of D7 retention compared to Control.

Q20. Using $\text{baseline}=34.21\%$, $\text{treatment}=44.91\%$ (effect size $h \approx 0.219$), the required sample size for 80% power at $\alpha=0.05$ is ≈ 326 users per group. The actual experiment used 1,213 and 1,287 — well over-powered for this effect size, meaning the significant result in Q9 is not a fragile, underpowered finding.

Q21. Required sample size scales roughly with $1/\text{effect}^2$, so halving the true lift would roughly *quadruple* the required sample size, not just double it — a key intuition for why detecting small effects is disproportionately expensive.

Q22-23 (Simpson's Paradox). New User segment: Flow A=95.0% vs Flow B=90.0% (A wins). Power User segment: Flow A=60.0% vs Flow B=50.0% (A wins again). Overall aggregated: Flow A=63.5% vs Flow B=86.0% — B wins overall. This is Simpson's Paradox: Flow A was shown mostly to the harder-to-convert Power User segment (900 of 1,000 exposures) while Flow B was shown mostly to the easier New User segment (900 of 1,000 exposures), so the segment mix — not the flow's real quality — dominates the naive aggregate.

Q24. The honest recommendation is Flow A — it wins in every actual segment, and the overall number is an artifact of unequal exposure, not a real quality difference. Propose a stratified or exposure-weighted comparison (or re-run with balanced/randomized allocation across segments) instead of reporting the raw overall rate.

Q25. Model answer approach: lead with the headline (Treatment lifted D7 retention from 34.2% to 44.9%, a statistically significant +31% relative improvement, more than adequately powered), note the platform nuance (iOS users retain somewhat better regardless of treatment, so consider whether the lift holds on both platforms), and flag that revenue impact couldn't be reliably modeled with the available predictors ($R^2 \approx 0.008$) — recommend a follow-up revenue-specific analysis (e.g., a two-part model) before making a monetization claim.

Q26. Candidates: `session_count_30d`, `avg_session_length_min`, `revenue_30d`, `platform`, or `nps_score` could all differ across groups by chance even with correct randomization. Check with an "A/A-style" covariate balance check: run a t-test/chi-square comparing each pre-treatment covariate (age, platform, country, `marketing_channel` — anything determined before the treatment could act) between Control and Treatment; large imbalances would suggest a broken randomization process rather than a true treatment effect.